Lesson 3.3 Guided Notes

Session 1 · Equivalent Ratios and Tables

I can build a ratio table by scaling both rows by the same factor. · 6.RP.A.3a

Word / Spanish	Meaning — complete the blank	Example
equivalent ratios / razones equivalentes	Ratios that make the same _____.	2 : 3 and 4 : 6
ratio table / tabla de razones	A table of _____ ratios. Each column is one pair.	3 : 2, 6 : 4, 9 : 6
scale factor / factor de escala	The number you _____ or divide both terms by.	2 : 3 × 4 = 8 : 12

Key idea

To find an equivalent ratio, multiply or divide _____ terms by the _____ number.

Adding the same number to both terms does _____ keep the ratio.

Worked example

A drink mix uses 5 scoops of powder for every 2 cups of water. How much water goes with 15 scoops?

Find the scale factor: 15 ÷ 5 = _____.

Multiply the water by the same factor: 2 × 3 = _____.

Check: 15 : 6 divided by 3 gives _____ again.

Answer: 15 scoops of powder go with _____ cups of water.

You try

1. Juice uses 1 cup of concentrate for every 4 cups of water. Complete the table.

_____

2. Fill in the missing number.
a. 5 : 2 = 20 : _____
b. 6 : 9 = 2 : _____

_____

Sum it up: To build a ratio table, I ___ both rows by the same number so that ___.

_____

Session 2 · The Bridge Strategy (Scaling Ratios)

I can use the bridge strategy to scale a ratio when I cannot multiply by a whole number. · 6.RP.A.3a

Word / Spanish	Meaning — complete the blank	Example
bridge ratio / razón puente	A _____ equivalent ratio you build first, then scale up.	8 : 20 = 2 : 5
simplify / simplificar	Divide both terms by the _____ number.	8 : 20 ÷ 4 = 2 : 5
greatest common factor / máximo común divisor	The _____ number that divides both terms.	GCF of 8 and 20 is 4

Key idea

When no whole number takes you to the target, _____ both terms down to a bridge ratio.

Then _____ both terms of the bridge up to the target.

Worked example

A trail mix uses 6 cups of nuts for every 8 cups of raisins. How many cups of raisins go with 15 cups of nuts?

6 × a whole number never equals _____.

Divide both terms by 2: 6 : 8 becomes _____.

Multiply both terms by 5: 3 : 4 becomes 15 : _____.

Answer: 15 cups of nuts go with _____ cups of raisins.

You try

1. A glaze uses 8 spoons of honey for every 20 spoons of water. How much water goes with 14 spoons of honey?

_____

2. Use a bridge to find the missing number.
a. 6 : 9 = 10 : _____
b. 12 : 16 = 9 : _____

_____

Sum it up: I use a bridge ratio when ___, and I check my answer by ___.

_____

Session 1 · Model, reason, and explain

Equivalent Ratios and Tables · 6.RP.A.3a

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A recipe uses 3 cups of oats for 2 cups of seeds. Complete the table for 6, 9, and 12 cups of oats. Explain why adding 3 to both entries of 3:2 does not preserve the recipe.

Oats (cups)	Seeds (cups)
3	2
6	?
9	?
12	?

_____

_____

_____

Extend your reasoning

Find an equivalent pair with less than 3 cups of oats.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

The Bridge Strategy (Scaling Ratios) · 6.RP.A.3a

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A mixture uses 6 cups of nuts for 8 cups of raisins. Find the raisins for 15 cups of nuts using a bridge through 3 cups and using a scale factor. Compare the methods.

Nuts	Raisins	Operation
6	8	start
3	4	÷ 2
15	?	× 5

_____

_____

_____

Extend your reasoning

Which method is more efficient for 1.5 cups of nuts? Justify a valid method.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?